

Mufan Li

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ACADEMIC POSITIONS

Assistant Professor, University of Waterloo
Department of Statistics and Actuarial Science 2025–Present

Postdoctoral Research Associate, Princeton University 2023–2025
Department of CSML and ORFE
Supervised by Boris Hanin

DEGREES

Ph.D. Statistics, University of Toronto 2017–2023
Thesis: *Analysis of Learning Algorithms via Diffusion Limits*
Supervised by Daniel M. Roy and Murat A. Erdogdu

M.Sc. Statistics, University of Toronto 2015–2016

B.A.Sc. Engineering Science, University of Toronto 2010–2015

PUBLISHED ARTICLES

See also my [Google Scholar](#) or [Semantic Scholar](#) pages.

1. Nolan Dey, Bin Claire Zhang, Lorenzo Noci, **M. Li**, Blake Bordelon, Shane Bergsma, Cengiz Pehlevan, Boris Hanin, and Joel Hestness, *Don't be lazy: CompleteP enables compute-efficient deep transformers*. NeurIPS (2025). [arXiv:2505.01618](#).
2. Sinho Chewi, Murat A. Erdogdu, **M. Li**, Ruoqi Shen, and Matthew Zhang, *Analysis of Langevin Monte Carlo from Poincaré to Log-Sobolev*. [Foundations of Computational Mathematics](#) (2024). COLT 2022 Extended Abstract. [arXiv:2112.12662](#).
3. Yunbum Kook, Matthew S. Zhang, Sinho Chewi, Murat A. Erdogdu, and **M. Li**, *Sampling from the Mean-Field Stationary Distribution*. COLT 2024. [arXiv:2402.07355](#).
4. **M. Li** and Mihai Nica, *Differential Equation Scaling Limits of Shaped and Unshaped Neural Networks*. TMLR 2024. [arXiv:2310.12079](#).
5. Blake Bordelon, Lorenzo Noci, **M. Li**, Boris Hanin, and Cengiz Pehlevan, *Depthwise Hyperparameter Transfer in Residual Networks: Dynamics and Scaling Limit*. ICLR 2024. M3L Workshop Oral Presentation. [arXiv:2309.16620](#).
6. Lorenzo Noci*, Chuning Li*, **M. Li***, Bobby He, Thomas Hofmann, Chris Maddison, and Daniel M. Roy, *The Shaped Transformer: Attention Models in the Infinite Depth-and-Width Limit*. NeurIPS 2023. [arXiv:2306.17759](#).
7. Matthew Zhang, Sinho Chewi, **M. Li**, Krishnakumar Balasubramanian, and Murat A. Erdogdu, *Improved Discretization Analysis for Underdamped Langevin Monte Carlo*. COLT 2023. [arXiv:2302.08049](#).
8. **M. Li** and Murat A. Erdogdu, *Riemannian Langevin Algorithm for Solving Semidefinite Programs*. Bernoulli (2023). [arXiv:2010.11176](#).
9. **M. Li**, Mihai Nica, and Daniel M. Roy, *The Neural Covariance SDE: Shaped Infinite Depth-and-Width Networks at Initialization*. NeurIPS 2022 (Selected for Oral, Nominated for Outstanding Paper Award). [arXiv:2206.02768](#).
10. Raphaël Berthier and **M. Li**, *Acceleration of Gossip Algorithms through the Euler–Poisson–Darboux Equation*. IMA Journal of Applied Mathematics (2022). [arXiv:2202.10742](#).

*Equal contribution.

11. **M. Li**, Mihai Nica, and Daniel M. Roy, *The Future is Log-Gaussian: ResNets and Their Infinite-Depth-and-Width Limit at Initialization*. [NeurIPS 2021](#). [arXiv:2106.04013](#).

PREPRINTS

1. **M. Li**, Jaume de Dios Pont, Mihai Nica, and Daniel M. Roy, *Geometric Dyson Brownian Motions and the Free Log-Normal Limit for a Non-Square Gaussian Matrix Product*. Preprint (2026). [arXiv:2606.30831](#).
2. Yihe Dong, Lorenzo Noci, Mikhail Khodak, and **M. Li**, *Is Random Attention Sufficient for Sequence Modeling? Disentangling Trainable Components in the Transformer*. Preprint (2025). [arXiv:2506.01115](#).
3. **M. Li**, and Maxime Gazeau, *Higher Order Generalization Error for First Order Discretization of Langevin Diffusion*. Preprint (2021). [arXiv:2102.06229](#).

RESEARCH FUNDING

NSERC Discovery Grant, Principal Investigator 2026–2031

AWARDS

Princeton DataX Postdoctoral Fellowship 2024–2025
 NSERC Postdoctoral Fellowship (Declined) 2024
 Doctoral Award, University of Toronto 2023
 Ontario Graduate Scholarship 2019–2023
 Student Research Presentation Award, Stat. Soc. of Canada 2021
 MITACS Accelerate Fellowship, with Borealis AI 2018–2019
 Undergraduate Summer Research Fellowship, University of Toronto 2012

RECENT INVITED TALKS

Statistics Seminar, Imperial College London June 2026
Feature Learning in the Proportional Depth-Width Limit
 ProbAI Theory of Scaling Laws Workshop, University of Warwick June 2026
Feature Learning in the Proportional Depth-Width Limit ([Video](#))
 Google DeepMind June 2026
Feature Learning in the Proportional Depth-Width Limit
 Math & Stats Colloquium, University of Guelph February 2026
The Proportional Scaling Limit of Neural Networks
 High Dimensional Learning Dynamics Workshop, BIRS February 2026
Depth Scaling of Feature Learning and Hyperparameter Transfer
 Southern Ontario Learning Theory (SOLT) Workshop November 2025
The Proportional Scaling Limit of Neural Networks
 Gatsby Seminar, University College London October 2025
The Proportional Scaling Limit of Neural Networks
 Stochastic Processes and Applications (SPA) Conference July 2025
The Proportional Scaling Limit of Neural Networks
 CRM Workshop: Random Matrices and High-Dim. Learning Dynamics June 2025
The Proportional Scaling Limit of Neural Networks
 Deep Learning Theory Seminar, University of Tokyo May 2025
The Proportional Scaling Limit of Neural Networks
 Math ML Seminar, MPI MiS and UCLA May 2025
The Proportional Scaling Limit of Neural Networks
 Probability Seminar, University of Washington February 2025
The Proportional Scaling Limit of Neural Networks

	INFORMS Annual Meeting <i>The Proportional Scaling Limit of Neural Networks</i>	October 2024
	STATQAM Seminar, UQAM <i>The Proportional Scaling Limit of Neural Networks</i>	September 2024
	Cerebras Systems <i>Infinite-Depth Neural Networks as Depthwise Stochastic Processes</i>	June 2024
	Transformers Seminar, Flatiron Institute <i>Neural Covariance SDE and the Shaped Transformer</i>	April 2024
	Alg-ML Seminar, Princeton University <i>Neural Covariance SDE and Its Limiting Spectrum</i>	April 2024
	One World Mathematics of ML Seminar <i>Infinite-Depth Neural Networks as Depthwise Stochastic Processes</i> (Video)	April 2024
EDITORIAL SERVICE	Action Editor, TMLR	2026–Present
	Area Chair, NeurIPS	2026
	Area Chair, ICML	2026
	Area Chair, ICLR	2024–2026
	Area Chair, ALT	2024–2026
REVIEWING	Journals AoAP, Bernoulli, FoCM, JCGS, JMLR, SICON, SIMODS, TMLR (Expert Reviewer)	
	Conferences COLT, ICLR, ICML, NeurIPS	
TEACHING	STAT 433/833 - Stochastic Processes 2	Fall 2026
	STAT 441 - Statistical Learning - Classification	Fall 2026
	STAT 946 - Deep Learning Theory	Fall 2025
	STAT 230 - Probability	Fall 2025
TEACHING MATERIALS	Lecture Notes on Neural Network Scaling Limits (PDF)	2026
MENTORING	Jonathan Borenstein, MMATH Stats, University of Waterloo	2025–2026
INDUSTRY EXPERIENCE	Research Intern, Borealis AI	Aug 2018–Apr 2019
	Investment Analyst, Ontario Teachers’ Pension Plan	Jul 2016–Jul 2017
	Electronic Trading Intern, RBC Capital Markets	May 2013–Aug 2014